

SEATTLE RENTON, WA, USA

WMO: 727934

Lat: 47.493N

Lon: 122.214W

Elev: 29

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94248

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	26.1	29.4	13.9	11.2	30.7	19.3	14.6	35.1	20.4	45.6	18.0	46.9	2.9	100	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.5	86.2	66.3	82.5	65.0	79.6	63.8	68.0	82.9	66.3	80.0	64.8	77.3	8.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	85.2	71.3	61.1	80.5	70.7	59.6	76.2	69.9	32.1	82.6	30.8	80.2	29.7	77.3	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	14.8	12.5	21.7	92.9	4.3	3.9	18.6	95.6	16.0	97.9	13.6	100.1	10.5	102.9
			20.2	71.1	4.4	4.2	17.0	74.1	14.5	76.5	12.0	78.8	8.8	81.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	42.5	43.6	47.0	51.2	57.3	62.4	67.5	67.8	62.3	53.7	46.4	41.7	(6)
	DBStd	10.55	5.71	4.97	4.81	4.79	5.24	5.13	4.58	4.12	4.57	4.68	5.84	5.86	(7)
	HDD50	996	238	184	118	39	0	0	0	0	19	134	261		(8)
	HDD65	4398	697	599	558	414	245	110	23	17	106	351	557	721	(9)
	CDD50	2339	6	5	24	75	230	372	543	552	368	133	27	4	(10)
	CDD65	270	0	0	0	0	6	32	101	105	25	1	0	0	(11)
	CDH74	2355	0	0	3	19	100	336	888	794	212	3	0	0	(12)
	CDH80	631	0	0	0	2	20	97	261	208	43	0	0	0	(13)
(14) Wind	WSAvg	5.3	4.9	5.3	5.9	5.9	5.7	5.9	5.9	5.4	4.9	4.5	4.8	4.8	(14)
(15) Precipitation	PrecAvg	48.7	6.8	4.4	5.2	4.1	3.0	2.4	1.0	1.2	2.0	4.7	7.3	6.3	(15)
	PrecMax	64.1	13.9	9.6	9.5	7.4	5.4	4.4	2.7	4.5	6.6	8.3	16.4	11.8	(16)
	PrecMin	37.4	3.1	0.0	1.7	1.2	0.8	0.8	0.0	0.2	0.1	1.4	2.7	2.6	(17)
	PrecStd	6.5	2.4	2.0	1.9	1.7	1.1	1.0	0.6	1.0	1.4	2.0	3.0	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	59.8	68.0	76.7	82.8	88.0	91.9	90.0	85.0	72.4	62.4	57.3	(19)
		MCWB	51.0	48.1	52.4	58.3	62.3	66.5	68.5	68.5	64.6	58.7	56.1	52.4	(20)
	2%	DB	54.8	55.5	62.0	68.4	76.5	82.5	86.3	85.2	79.9	67.3	58.7	54.1	(21)
		MCWB	50.1	48.2	50.6	54.5	60.6	64.2	66.6	66.6	63.7	57.6	53.7	49.0	(22)
	5%	DB	52.5	53.5	57.6	63.7	72.0	77.4	82.3	81.7	75.3	63.9	56.3	51.9	(23)
		MCWB	48.4	47.5	49.0	52.5	58.7	61.8	65.1	65.4	62.0	56.2	51.9	47.6	(24)
	10%	DB	50.3	51.5	54.7	60.2	67.6	72.9	79.1	78.7	72.1	61.4	54.3	49.7	(25)
		MCWB	46.8	46.3	47.9	50.7	56.7	60.0	63.9	64.2	60.8	55.1	50.4	45.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	52.6	55.3	59.5	64.0	67.9	70.5	69.8	67.7	62.4	57.7	54.3	(27)
		MCDB	56.4	56.0	63.7	73.8	78.5	85.2	86.8	85.0	79.6	66.9	60.7	56.9	(28)
	2%	WB	51.8	50.7	52.8	55.8	61.7	65.1	67.9	68.1	64.9	59.6	55.1	50.7	(29)
		MCDB	53.6	53.8	58.6	65.7	74.6	79.8	83.4	82.5	76.0	64.3	58.0	52.6	(30)
	5%	WB	49.2	48.5	50.8	53.7	59.4	62.9	66.1	66.4	63.3	57.7	53.1	48.2	(31)
		MCDB	51.2	51.7	56.2	61.0	70.1	75.4	80.7	79.4	73.0	62.0	55.4	50.6	(32)
	10%	WB	47.1	46.7	49.1	51.7	57.4	60.9	64.4	65.0	61.8	56.1	51.1	46.3	(33)
		MCDB	49.4	50.0	54.1	57.9	65.9	71.8	77.6	76.7	70.1	60.5	53.5	48.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	12.1	13.6	15.0	16.0	16.3	18.6	18.5	16.9	13.3	11.2	9.3	(35)
		MCDBR	11.9	15.5	18.5	22.4	24.7	25.2	25.8	25.1	24.5	17.7	12.3	10.3	(36)
	5% WB	MCWBR	7.8	9.0	10.0	10.9	11.4	10.6	10.3	10.2	10.9	9.9	8.0	7.4	(37)
		MCWBR	9.7	11.2	15.4	19.2	21.7	22.6	24.1	22.4	20.9	14.2	10.5	9.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.328	0.320	0.328	0.363	0.381	0.371	0.356	0.362	0.350	0.333	0.323	0.334	(40)
		taud	2.481	2.538	2.536	2.405	2.377	2.431	2.490	2.482	2.507	2.557	2.534	2.388	(41)
	Ebn at Noon		239	271	285	283	280	283	286	280	273	262	240	217	(42)
		Edh at Noon	21	24	28	35	37	35	33	32	28	23	20	20	(43)
(44) All-Sky Solar Radiation	RadAvg		326	605	910	1339	1682	1855	2075	1767	1283	730	381	264	(44)
	RadStd		46	79	102	123	135	176	150	116	112	80	50	38	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	+0.95	N/A	N/A	N/A	+238	+79
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page